



Start with ifcSQL_Tools QGIS plugin

Main developer: Michele Berlato

Contributors: Giorgio Agugiaro, Ken Arroyo Ohori, Carlo Zanchetta



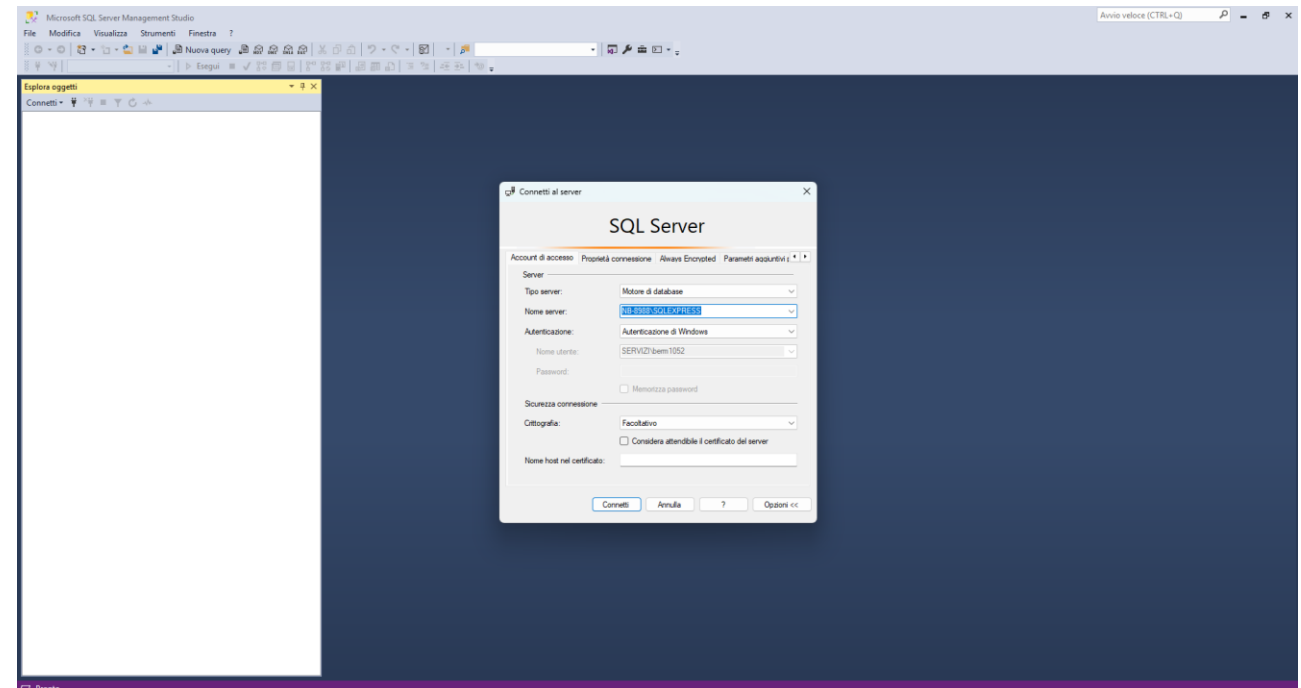
MSSQL



Download and install Microsoft SQL server (MSSQL) and SQL Server Management Studio (SSMS)

- Microsoft SQL Server Express 22
(<https://www.microsoft.com/it-it/download/details.aspx?id=104781>)
- SQL Server Management Studio 20
(<https://learn.microsoft.com/it-it/ssms/release-notes-20>)

It is important to download and install these versions.





PostgreSQL

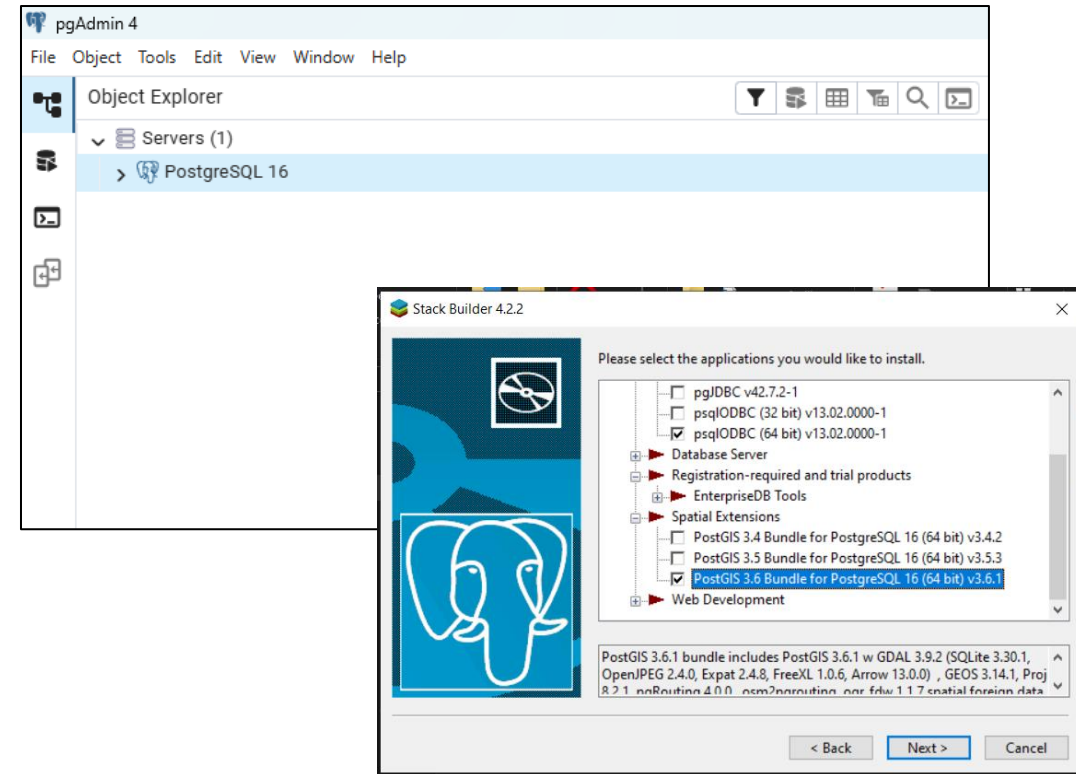


Download and install PostgreSQL and pgAdmin

You can download and install everything together:

<https://www.enterprisedb.com/downloads/postgres-postgresql-downloads>

Remember to activate PostGIS during the installation.

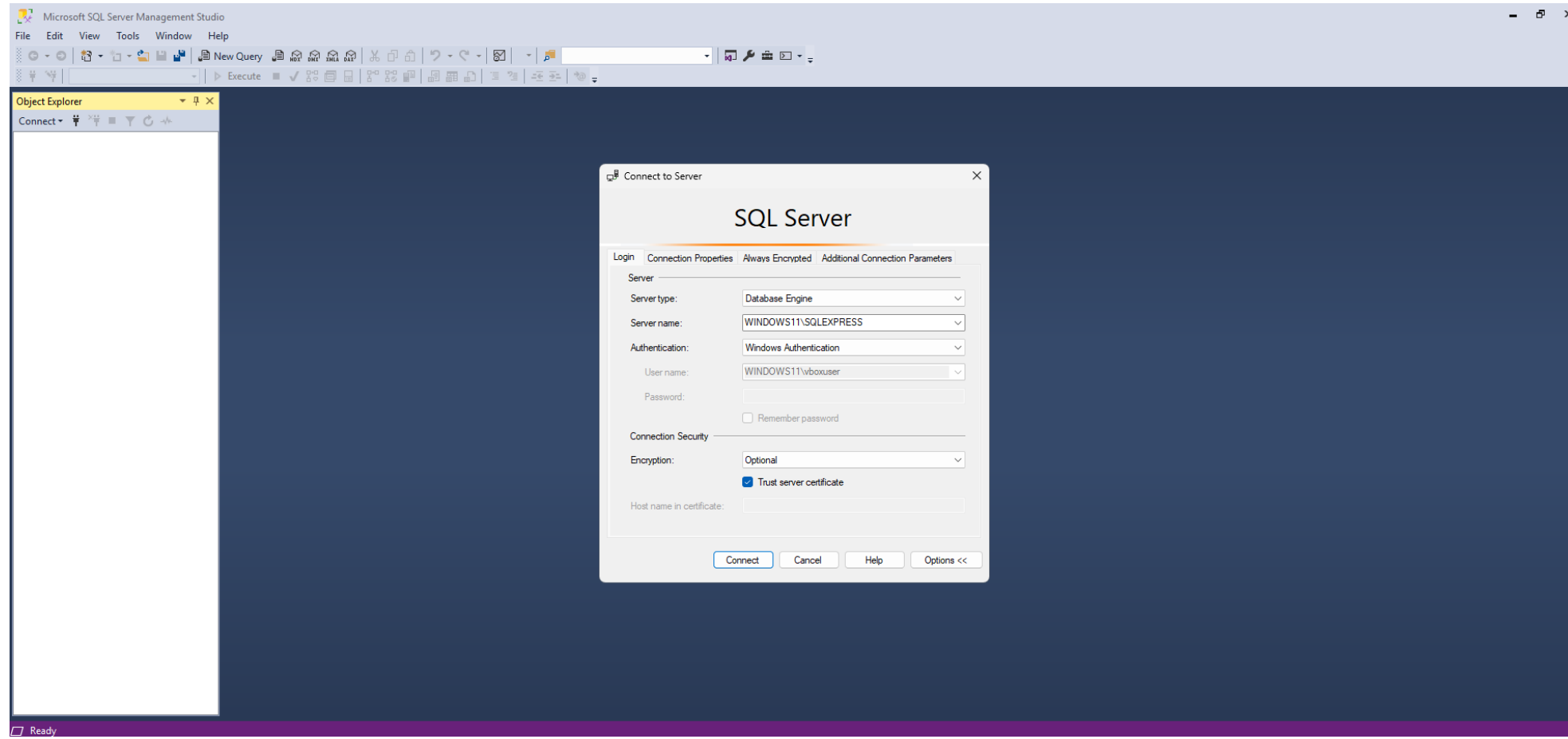




Connect to MSSQL



Proceed with Windows authentication.

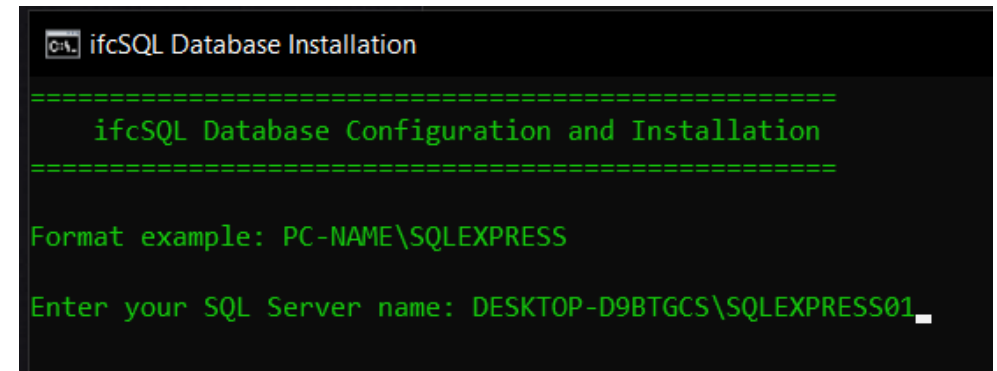
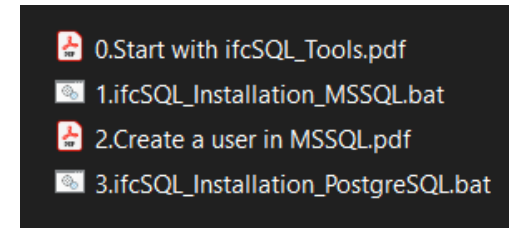




Click the “1.ifcSQL_Installation_MSSQL.bat” file to install ifcSQL on MSSQL

Press a key until the process is finish.

The process may take some minutes.





ifcSQL_Installation_MSSQL



Click on the bat file.

1

```
ifcSQL Database Installation

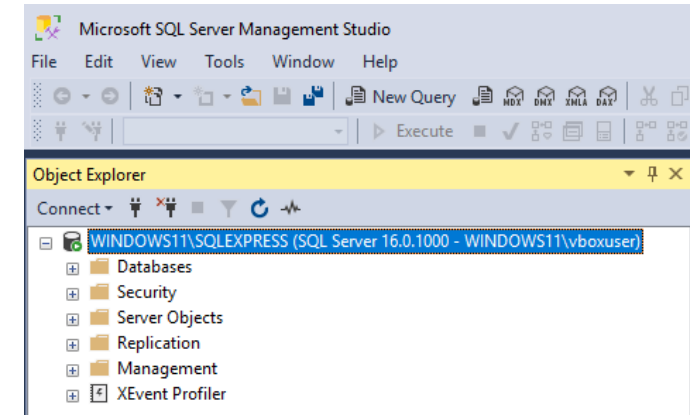
=====
ifcSQL Database Configuration and Installation
=====

Format example: PC-NAME\SQLEXPRESS

Enter your SQL Server name:
```

Retrieve the server name from MSSQL.

2



Paste the server name.

3

```
ifcSQL Database Installation

=====
ifcSQL Database Configuration and Installation
=====

Format example: PC-NAME\SQLEXPRESS

Enter your SQL Server name: Windows11\SQLEXPRESS|
```

Press a key until the process is finished.

4

```
ifcSQL Database Installation

=====
ifcSQL Database Configuration and Installation
=====

Format example: PC-NAME\SQLEXPRESS

Enter your SQL Server name: Windows11\SQLEXPRESS

[INFO] Testing connection to 'Windows11\SQLEXPRESS'...
[SUCCESS] Connected successfully to the SQL Server!

=====
Server set to: Windows11\SQLEXPRESS
Starting the procedure...

Note: The procedure will pause several times to show progress.
Press "Enter" whenever prompted to continue.

=====
--- PHASE 1: Database Creation ---
Press any key to continue . . .
```



MSSQL ifcSQL database



You should see that the database has been successfully created.

The screenshot shows the Microsoft SQL Server Management Studio interface. The Object Explorer on the left displays the database structure, including tables and views. The central pane shows a SQL query executed against the ifcSQL database. The query selects various fields from the ifcProject table. The bottom pane displays the results of the query, showing a table with 7 rows and 12 columns.

```
USE [ifcSQL]
GO

SELECT [ProjectId]
,[ProjectName]
,[ProjectDescription]
,[ProjectGroupId]
,[SpecificationId]
,[Author]
,[Organization]
,[OriginatingSystem]
,[Documentation]
,[ParentProjectId]
,[ProjectTypeId]
FROM [ifcProject].[Project]
GO
```

ProjectId	ProjectName	ProjectDescription	ProjectGroupId	SpecificationId	Author	Organization	OriginatingSystem	Documentation	ParentProjectId	ProjectTypeId
0	-	-	0	40	-	-	-	-	0	0
1001	IFC2x3	IFC2x3 Schema	1	23	NULL	NULL	NULL	NULL	0	0
1002	IFC4	IFC4 Schema	1	40	NULL	NULL	NULL	NULL	0	0
1003	IFC4x1	IFC4x1 Schema	1	41	NULL	NULL	NULL	NULL	0	0
1004	IFC4x2	IFC4x2 Schema	1	42	NULL	NULL	NULL	NULL	0	0
1005	IFC4x3	IFC4x3 Schema	1	43	NULL	NULL	NULL	NULL	0	0
1006	hello_project_output	IFC4	4	40	Bernhard Simon Bock, Friedrich Eder	https://github.com/IfcSharp			0	0

Query executed successfully. WINDOWS11\SQLEXPRESS (16.0 ... WINDOWS11\vboxuser (79) ifcSQL 00:00:00 7 rows



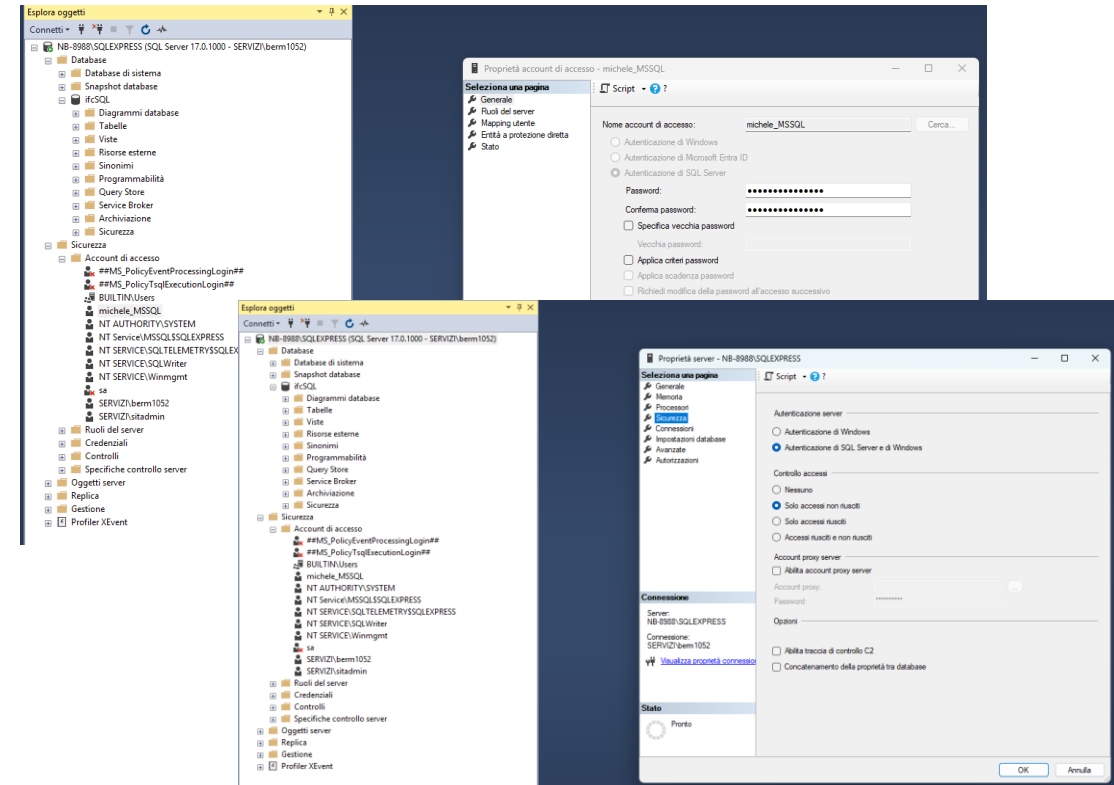
Create a USER in MSSQL



Follow the “**2.Create a user in MSSQL**” guideline.

Summary of the step:

- Create an MSSQL user with SQL server authentication and password.
- Give the user administrator privileges.
- Set up access with mixed authentication (SQL server + Windows).
- Verify that the user can log in to MSSQL.





Click the “3.ifcSQL_Installation_PostgreSQL.bat” file to install ifcSQL on PostgreSQL

REMEBER: run as administrator the bat file.

Follow the instruction in the bat.

Check at the end to make sure everything is working; you may have entered the wrong credentials.

```
Administrator: ifcSQL Database Setup and TDS FDW Installation
=====
PostgreSQL 16 Database Setup & FDW Installation
=====
[INFO] PostgreSQL base path found at: C:\Program Files\PostgreSQL\16
[INFO] Deploying TDS FDW files to PostgreSQL directories...
Copying extension files (.control, .sql)...
Updating module_pathname in tds_fdw.control...
Stopping PostgreSQL service to unlock DLL files...
Copying DLL binaries to bin folder...
Copying tds_fdw.dll to lib folder...
Restarting PostgreSQL service...
[SUCCESS] FDW files deployed successfully!

--- Database Creation ---
Enter PostgreSQL username (e.g., postgres):
```

```
--- Connection Test ---
Attempting to read the first 5 rows from 'ifcinstance.entity'...
ERROR: DB-Library error: DB #: 20009, DB Msg: Unable to connect: Adaptive Server is unavailable or does not exist (Windows11\SQLEXPRES), OS #: 0, OS Msg: No error, Level: 9

[ERROR] Could not read from MSSQL. Check if the MSSQL server is reachable and credentials are correct (server name; port; username; password).

Do you want to re-enter the MSSQL credentials and try again? (Y/N): y
```



Ready to start



You are ready to start using the plugin!
(find out how to use it on GitHub)



Bonus



Would you like to set up a second or new database system (MSSQL + PostgreSQL)?

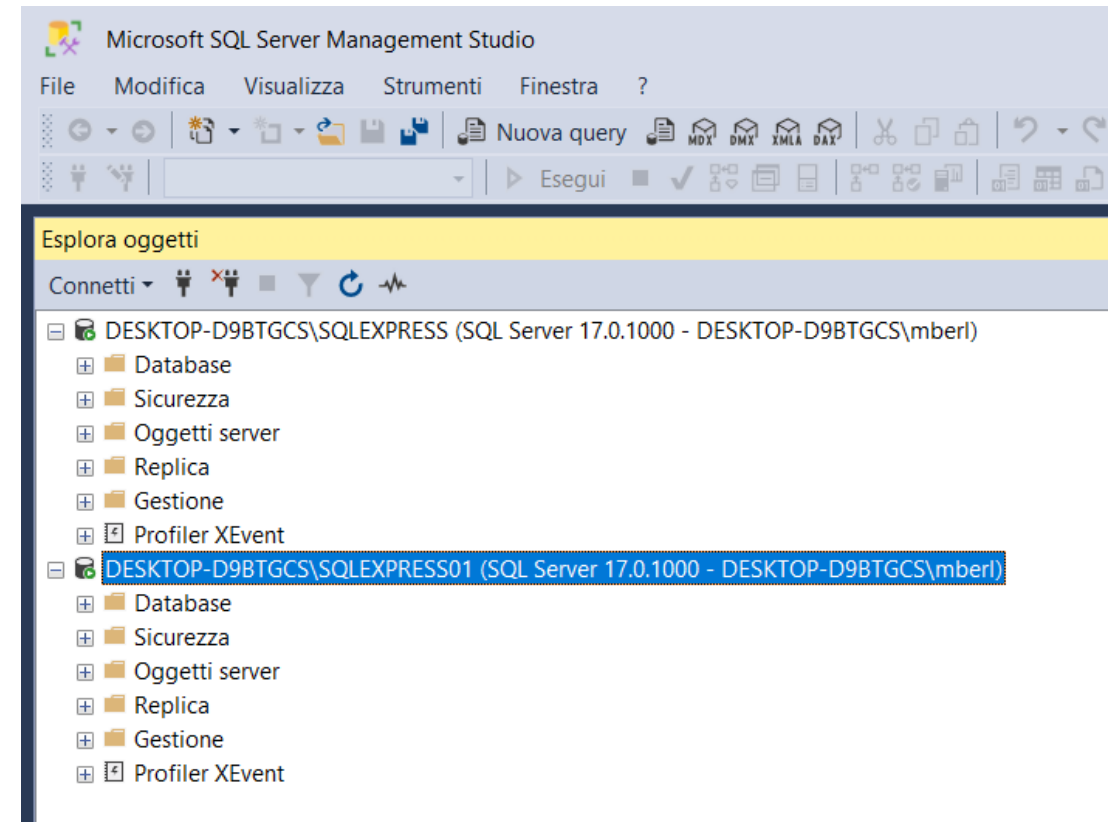


MSSQL



Install **again** Microsoft SQL server (MSSQL)

- Microsoft SQL Server Express 22
(<https://www.microsoft.com/it-it/download/details.aspx?id=104781>)
- A second database will be installed.





Click the “**1.ifcSQL_Installation_MSSQL.bat**” file to install ifcSQL on MSSQL

This time, use the SQL Server Name of the **second database**.

```
ifcSQL Database Installation

=====
  ifcSQL Database Configuration and Installation
=====

Format example: PC-NAME\SQLEXPRESS

Enter your SQL Server name: DESKTOP-D9BTGCS\SQLEXPRESS01_
```




Click the “3.ifcSQL_Installation_PostgreSQL.bat” file to install ifcSQL on PostgreSQL

REMEBER: run as administrator the bat file.

Answer “Y” to the question “Do you want to create a new numbered database?”

```
Administrator: ifcSQL Database Setup and TDS FDW Installation
=====
PostgreSQL 16 Database Setup & FDW Installation
=====

[INFO] PostgreSQL base path found at: C:\Program Files\PostgreSQL\16

[INFO] Deploying TDS FDW files to PostgreSQL directories...
Copying extension files (.control, .sql)...
Updating module_pathname in tds_fdw.control...
Stopping PostgreSQL service to unlock DLL files...
Copying DLL binaries to bin folder...
Copying tds_fdw.dll to lib folder...
Restarting PostgreSQL service...
[SUCCESS] FDW files deployed successfully!

--- Database Creation ---
Enter PostgreSQL username (e.g., postgres): postgres
Enter password for user postgres: 5432

[INFO] The database 'ifcSQL' already exists.
Do you want to create a new numbered database? (Y/N): _
```



The end



You can repeat the procedure as many times as you want.